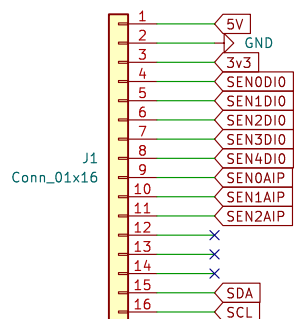
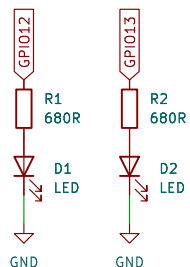


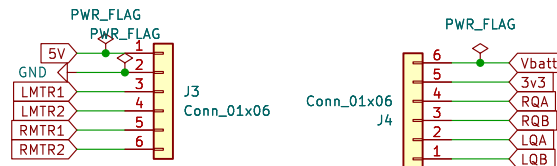
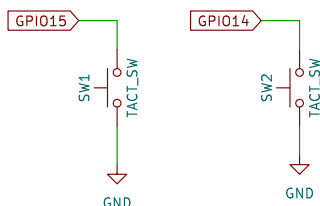
Sensor connector



Board LEDs



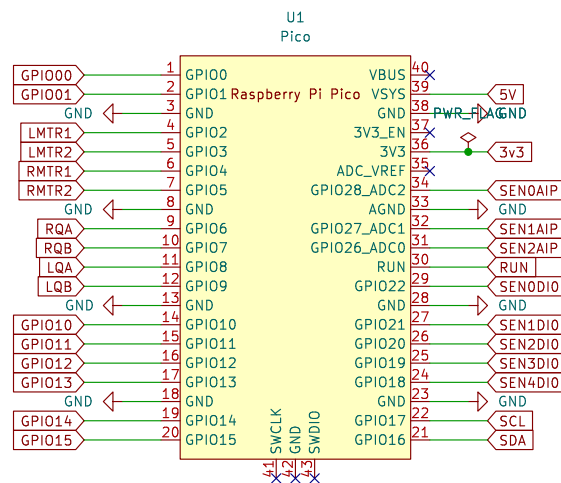
Board switches/push buttons



Mezzanine connections

MicroController voltage is 3.3v for any RP2040

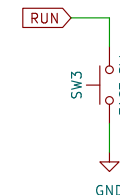
Raspberry Pi Pico / PicoW



MECHANICAL

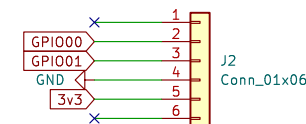


PiPico RESET push button

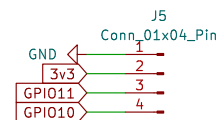


Breakout Connectors

J2 presents GP0 and GP1 with 3v3 and Gnd on pins suitable for bluetooth module to communicate on UART0. The pins can also present I2C0
A vs838 IR receiver could be installed on J2.



J5 presents GP10 and GP11 for I/O or alternative connection to I2C1 suitable for connecting 0.9" OLED screen or KY-022 IR receiver



UKMARS Gemini RP2040 Mezzanine Board Schematic

Sheet: /

File: UKMARSBOT_RP2040Mezzanine.kicad_sch

Title: UKMARS Gemini RP2040 Mezzanine V2.0

Size: A4

Date: 2025-07-07

Rev: 2

KiCad E.D.A. 8.0.8

Id: 1/1